

Yangxinyu Xie

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Education

- 2022 – 2026 **Ph.D.**, Wharton School, University of Pennsylvania, *Statistics and Data Science*.
Advisors: Linda H. Zhao, Weijie J. Su
Thesis: *Statistical Gems in AI: Toward Reliable and Efficient Intelligence*
- 2024 – 2025 **Visiting Student**, *Mathematics and Computer Science*, Argonne National Laboratory.
Advisor: Tanwi Mallick
- 2021 – 2022 **Master of Science**, University of Texas at Austin, *Computer Science*.
Advisors: Anna Gál and Ngoc Mai Tran
Thesis: *Matrix Rigidity: A Survey*
- 2017 – 2021 **Bachelor of Science**, University of Texas at Austin, *Mathematics, Computer Science*.

Professional Experience

- 2026 – **Quantitative Researcher**, *Jump Trading*, Chicago, IL.
- Summer 2024 **Givens Associate**, *Argonne National Laboratory*, Lemont, IL.

Publications

- Preprint
- Shomee, H. H., Chaturvedi, R., Xie, Y., and Mallick, T. SCORE: Specificity, Context Utilization, Robustness, and Relevance for Reference-Free LLM Evaluation. arXiv preprint arXiv:2602.10017.
 - Alcala, J., Andreeva, R., Kobzar, V. A., Mixon, D. G., Na, S., Sule, S., and Xie, Y. Neural Collapse in the Orthoplex Regime. arXiv preprint arXiv:2603.20587.
 - Xie, Y. and Bhattacharya, B.B. Joint Poisson Convergence of Monochromatic Hyperedges in Multiplex Hypergraphs. arXiv preprint arXiv:2412.00610.
 - Baptista, R., O'Reilly, E. and Xie, Y. TrIM: Transformed Iterative Mondrian Forests for Gradient-based Dimension Reduction and High-Dimensional Regression. arXiv preprint arXiv:2407.09964.
 - Dhruv, A., Xie, Y., Branham, J., and Mallick, T. Comparative Evaluation of Prompting and Fine-Tuning for Applying Large Language Models to Grid-Structured Geospatial Data. arXiv preprint arXiv:2505.17116.
- 2026
- Chen, X., Li, X., Xie, Y., and Long, Q. Selective Disclosure Watermarking for Large Language Models. Accepted at International Conference on Machine Learning (ICML 2026).
 - Xie, Y., Wang, T., Mallick, S., Sun, Y., Noarov, G., Yu, M., Mallick, T., and Dobriban, E. Statistical Early Stopping for Reasoning Models. Accepted at International Conference on Machine Learning (ICML 2026).
 - Xie, Y., Chen, X., Ren, Z., and Su, W. J. (2026). Watermark in the Classroom: A Conformal Framework for Adaptive AI Usage Detection. Harvard Data Science Review, 8(1).

- 2025
- Tolnai, B.A., Zimmermann, R.S., Xie, Y., Tran, N., Çeliker, C.E., Ma, Z.G., Sartori, I., Amos, M., Bengtsson, G., Lien, S.K. and Miller, C. Advancing non-intrusive load monitoring: insights from the winning algorithms in the ADRENALIN 2024 load disaggregation competition. In Nordic Energy Informatics Academy Conference (pp. 338-354).
 - Noarov, G., Mallick, S., Wang, T., Joshi, S., Sun, Y., Xie, Y., Yu, M. and Dobriban, E. Foundations of Top- k Decoding for Language Models. In Advances in Neural Information Processing Systems 38 (NeurIPS 2025).
 - de Rijke, M., van den Hurk, B., Salim, F., Al Khouardajie, A., Bai, N., Calzone, R., Curran, D., Demil, G., Frew, L., Gießing, N., Gupta, M. K., Heuss, M., Hobeichi, S., Huard, D., Kang, J., Lucic, A., Mallick, T., Nath, S., Okem, A., Pernici, B., Rajapakse, T., Saleem, H., Scells, H., Schneider, N., Spina, D., Tian, Y., Totin, E., Trotman, A., Valavandan, R., Workneh, D., Xie, Y. *Report on the 1st Workshop on Information Retrieval for Climate Impact (MANILA24) at SIGIR 2024*. In ACM SIGIR Forum (Vol. 59, No. 1, pp. 1-23). New York, NY, USA: ACM.
 - Xie, Y., Li, X., Mallick, T., Su, W. J. and Zhang, R. Debiasing Watermarks for Large Language Models via Maximal Coupling. *Journal of the American Statistical Association*, 120(551), pp. 1424–1436.
 - Xie, Y., Jiang, B., Mallick, T., Bergerson, J., Hutchison, J.K., Verner, D.R., Branham, J., Alexander, M.R., Ross, R.B., Feng, Y. and Levy, L.A. MARSHA: multi-agent RAG system for hazard adaptation. *npj Climate Action*, 4(1), p.70.
 - Jiang, B., Xie, Y., Wang, X., Yuan, Y., Hao, Z., Bai, X., Su, W. J., Taylor, C. J., and Mallick, T. Towards Rationality in Language and Multimodal Agents: A Survey. In Proceedings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL).
- 2024
- Jiang, B., Xie, Y., Hao, Z., Wang, X., Taylor, C. J., Su, W. J., Mallick, T. and Roth, D. A Peek into Token Bias: Large Language Models Are Not Yet Genuine Reasoners. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP).
- 2023
- Krenn, M., Buffoni, L., Coutinho, B., Eppel, S., Foster, J.G., Gritsevskiy, A., Lee, H., Lu, Y., Moutinho, J.P., Sanjabi, N., Sonthalia, R., Tran, N., Valente, F., Xie, Y., Yu, R., and Kopp, M. Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially growing knowledge network. *Nature Machine Intelligence*, pp.1-10. <https://doi.org/10.1038/s42256-023-00735-0>
- 2022
- Sun, A.Y., Jiang, P., Yang, Z.L., Xie, Y. and Chen, X. A graph neural network (GNN) approach to basin-scale river network learning: the role of physics-based connectivity and data fusion. *Hydrology and Earth System Sciences*, 26(19), pp.5163-5184.
- 2021
- Tran NM, Xie, Y. Improving random walk rankings with feature selection and imputation. In 2021 IEEE International Conference on Big Data (Big Data) 2021 Dec 15 (pp. 5824-5827). IEEE.
 - Xie, Y., Kulpanowski, D., Ong, J., Nikolova, E., Tran, NM. Predicting Covid-19 emergency medical service incidents from daily hospitalization trends. *Int J Clin Pract*. 2021; 00:e14920. doi:10.1111/ijcp.14920
 - Xie, Y. On 2×2 Tropical Commuting Matrices. *Linear Algebra and its Applications*, 620, 92-108.

Public Scholarship

- 2024 ○ Xie, Y., Pizzurro, J., Santosa, F. and O'Reilly, E., 2024. An assessment of the impact of the Baltimore Red Line on job accessibility. Press release, Johns Hopkins University, 7 June. Available at: <https://hub.jhu.edu/2024/06/07/red-line-job-access-report/>

Competitions and Awards

- 2024 **Load Disaggregation Challenge: Energy Use in Buildings, 2nd Place, ADRENALIN Energy Competitions.**
Team Members: Ngoc Mai Tran, Yangxinyu Xie
- 2023 **CityLearn Challenge Forecasting Track, 2nd Place, NeurIPS.**
Team Members: Tanwi Mallick, Ngoc Mai Tran, Yangxinyu Xie
- 2022 **Longhorn Research Poster Session Award, 3rd Place, Research Week, University of Texas at Austin.**
- 2021 **Science4cast Competition, 2nd Place, IEEE BigData Cup Challenges, Institute of Advanced Research in Artificial Intelligence, Vienna, Austria.**
Team Members: Ngoc Mai Tran, Yangxinyu Xie
- 2021 **Smart City Hackathon, 1st Place, Good Systems, University of Texas at Austin.**
Team Members: Jacob Helwig, Jack Si, Yangxinyu Xie

Scholarships, Fellowships and Grants

- 2025 **Mack Institute PhD Research Fellowship, The Wharton School.**
- 2025 **Travel Award by George James Term Fund, Wharton Doctoral Programs.**
- 2024 **AMS Graduate Student Travel Grants for Joint Mathematics Meeting.**
- 2023 **Sustainable Research Pathways, Sustainable Horizons Institute & U.S. Department of Energy Exascale Computing Project.**
- 2022 **Boyd Harshburger Travel Award, The Southern Regional Council on Statistics.**
- 2021 **Graduate School Application Financial Aid Program, oSTEM, Queer in AI.**
- 2021 **Student Fellow, College of Undergraduate Studies, University of Texas at Austin.**
- 2021, 2020 **AMS Undergraduate Student Travel Grants for Joint Mathematics Meeting.**

Presentations

Research Talks

- 2025 ○ Debiasing Watermarks for Large Language Models via Importance Sampling. AMS Special Session on Trustworthy AI Applications Including Machine Learning, PINN, and Inverse Problems. Joint Mathematics Meetings. Seattle, WA, January, 2025.
- Unveiling Large Language Model Vulnerabilities: A Geometric and Topological Approach to Adversarial Robustness. AMS Special Session on MRC Mathematics of Adversarial, Interpretable, and Explainable AI. Joint Mathematics Meetings. Seattle, WA, January, 2025.

- 2024
 - Statistical Gems in AI Safety. Data-driven Decision-making: Personalization, Practicality, Perspicuity. INFORMS Annual Meeting. Seattle, WA, October, 2024.
 - WildfireGPT: Tailored Large Language Model for Wildfire Analysis. The SIGIR 2024 Workshop on Information Retrieval for Climate Impact. Washington D.C. July, 2024.
 - Unbiased Watermark for Large Language Models via Importance Sampling. Lawrence D. Brown Student Workshop. University of Pennsylvania. March, 2024.
 - Joint Convergence of Monochromatic Edges in Multiplex Hypergraphs. UNL Discrete Seminar (DisCo). University of Nebraska-Lincoln. February, 2024.
 - Joint Convergence of Monochromatic Edges in Multiplex Hypergraphs. AMS Contributed Paper Session on Combinatorics, IV. Joint Mathematics Meetings. San Francisco, CA, January, 2024.
- 2023
 - A Comparative Study of Loss Functions: Traffic Predictions in Regular and Congestion Scenarios. Summer Argonne Student Symposium (Sassy). Virtual, August, 2023.
- 2021
 - Random walk rankings with feature selection and imputation. IEEE International Conference on Big Data, Virtual, December, 2021.
 - Optimizing Ambulance Allocation and Routing During Extreme Events. Smart Cities Consortium, Good Systems, University of Texas at Austin, Austin, TX, January, 2021.
- 2020
 - Matrix Rigidity, Algebraic and Combinatorial Techniques. 3rd Bringing Young Mathematicians Together Conference, Universitat de Valencia, Virtual, December, 2020.
- 2019
 - On 2 by 2 Tropical Commuting Matrices. Southeastern Undergraduate Mathematics Workshop, Georgia Institute of Technology, Atlanta, GA, August, 2019.

Poster Presentations

- 2024
 - WildfireGPT: Tailored Large Language Model for Wildfire Analysis. Tackling Climate Change with Machine Learning Workshop, NeurIPS, Vancouver, BC, December, 2024.
 - A Peek into Token Bias: Large Language Models Are Not Yet Genuine Reasoners. Workshop on Statistical Frontiers in LLMs and Foundation Models, NeurIPS, Vancouver, BC, December, 2024.
- 2022
 - Towards Label Driven Mondrian Forests. The Southern Regional Council on Statistics, 57th Summer Research Conference, Jekyll Island, GA, October, 2022.
 - Predicting Covid-19 EMS Incidents from Daily Hospitalization Trends. Longhorn Research Poster Session, UT Austin, April, 2022. *Poster Presentation Award, Third Place.*
- 2021
 - Optimizing EMS Responses during Extreme Events. MAA Undergraduate Poster Session, Joint Mathematics Meetings, January, 2021. *Honorable Mention Poster.*
- 2020
 - On Tropical Commuting Matrices. MAA Undergraduate Poster Session, Joint Mathematics Meetings, Denver, CO, January, 2020.

Public Scholarship

- 2025
 - A Data-Driven Analysis of the Surge in Anti-Transgender Bills in U.S. State Legislatures, 2018-2023. Spectra Special Session. Joint Mathematics Meetings. Seattle, WA, January, 2025.

- 2024 ○ Personalized and Context-specific Climate Resilience and Adaptation With Large Language Model, GIS DAY 2024, University of Pennsylvania, November 2024.
- 2023 ○ Modeling the Impact of the Baltimore Red Line on Job Accessibility, GIS DAY 2023, University of Pennsylvania, November 2023.
- All Aboard! Assessing Transit Equity in Baltimore, MD, Impact Summit, August 2023.

Professional Service

Reviewer

- Journals ○ *Stat*
- Conferences ○ ACL Rolling Review (ARR) 2026
- Workshops ○ ICML 2025 Workshop on Multi-Agent Systems in the Era of Foundation Models
- NLP for Positive Impact Workshop at EMNLP 2024
- NeurIPS 2024 Workshop on Behavioral ML

Conference Organization

- Workshop ○ Program committee, MANILA25: Information Retrieval for Climate Impact, SIGIR 2025
- Symposium ○ Organizing Committee, Wharton Innovation Doctoral Symposium (WINDS) 2025

Recognition and Distinctions

- 2022 **Dean's Honored Graduate**, *College of Natural Sciences*, University of Texas at Austin.
- 2022 **Research Distinction**, *College of Natural Sciences*, University of Texas at Austin.
- 2021 **UT Austin Member**, *Academy of Undergraduate Researchers Across Texas*.
- 2021 **George H. Mitchell Student Awards Semi-Finalist**, *Student Success Initiatives*, University of Texas at Austin.
- 2021 **Distinguished College Scholar**, *College of Natural Sciences*, University of Texas at Austin.
- 2021 **Cactus Standout Award**, *Cactus Yearbook*, University of Texas at Austin.
- 2019 **Second Year Excellence Award**, *College of Natural Sciences*, University of Texas at Austin.
- Fall 2017 **Dean's List**, *College of Undergraduate Studies*, University of Texas at Austin.

Professional Development

- Summer 2024 **Mathematics Research Communities (MRC)**, *Mathematics of Adversarial, Interpretable, and Explainable AI*, American Mathematical Society.

Teaching

Certification

- Spring 2024 **The Carpentries Coding Instructor**, *Certified by: Karen Word*.

Workshops

- May 2025 **Data Carpentries: Introduction to Python**, *Center for Undergraduate Research and Fellowships*, w/ Ashay Katre, Ruben Ramirez Salas and Hao Ye, University of Pennsylvania Libraries.
- April 2025 **From Data to Discovery: Exploring AI with a Patent Case Study, ChatGPT, and Generative Models**, *Women in Data Science @ Penn (WiDS @ Penn) Conference*, w/ Linda Zhao, University of Pennsylvania.
- March 2025 **Data Carpentries: Python for Social Scientists**, *Research Data & Digital Scholarship*, w/ Ashay Katre, University of Pennsylvania Libraries.
- March 2024 **Data Carpentries: Python for Social Scientists**, *Research Data & Digital Scholarship*, w/ Ruben Ramirez Salas and Hao Ye, University of Pennsylvania Libraries.
- [Teaching Assistantship - University of Pennsylvania](#)
- Fall 2024 **Introductory Statistics**, *Undergraduate*.
- Fall 2024 **Stat Computing with R**, *Undergraduate*.
- Fall 2023 **Statistical Methodology**, *Graduate*.
- Fall 2022 **Applied Econometrics I**, *Graduate*.
- [Teaching Assistantship -University of Texas at Austin](#)
- Spring 2020 **Algorithms and Complexity: Honors**, *Undergraduate*.
- Spring 2020 **Predictive Analytics**, *Undergraduate*.
- Fall 2019 **Applied Regression and Time Series**, *Undergraduate*.
- Fall 2019 **Discrete Mathematics**, *Undergraduate*.

--- [Mentorship](#)

- Spring 2025 **PhD Student Leader**, *Summit@Wharton*, University of Pennsylvania.
- Spring 2023 **Graduate Student Mentor**, *Undergraduate Research Program in Probability and Statistics*, University of Pennsylvania.
- Fall 2022 **Graduate Student Mentor and Panelist**, *3rd annual Statistics Undergraduate Research Experience (SURE)*, The Southern Regional Council on Statistics.
- Spring 2021 – **Mentor**, *Directed Reading Program, ACM For Change*, University of Texas at Austin.
- Spring 2022

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- 2024 – 2025 **Board Member**, *Independence Business Alliance*, LGBTQ+ Chamber of Commerce for Greater Philadelphia.
- Fall 2022 – **Council Member**, *Wharton Doctoral Council*, University of Pennsylvania.
- 2025
- 2023 **Participant**, *Michael S. Hinson, Jr. LGBTQ+ Leadership Pipeline Program*, Philadelphia Office of LGBT Affairs.
- Spring 2023 **Volunteer**, *Community Engagement Program*, University of Pennsylvania.

--- [Skills](#)

Fluent Python, R, $\text{\LaTeX}$

Basic HTML, Shell, Java, C#, C, C++, SQL